PROBLEM 06

Particle on Sinusoidal Curve

Original statement · Problems.pdf, p. 7

Frenet-Serret Frame

A point particle P travels along a curve given by:

$$y = \sin x$$

with a velocity along x given by $\dot{x} = 1$.

(a) Find the intrinsic, unitary vector basis associated to the curve.

(b) Express x as a function of an arc length parameter, s. You can leave your solution as a function of an integral.

(c) Find the tangent and normal acceleration components of P as a function of x.

(d) Write the radius of curvature of the trajectory as a function of x.

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle P travels along the curve $y = \sin x$ with horizontal velocity $\dot{x} = 1$.
- **Hypotheses:** the data are in SI units, so $y = \sin x$ means $y = (1\ \mathrm{m})\sin(x/1\ \mathrm{m})$ and $\dot{x} = 1\ \mathrm{m/s}$. We take $x(0) = 0$, so $x(t) = t$ (in these units).
- **Constraints and degrees of freedom:** the curve $y = \sin x$ is a holonomic, scleronomic constraint in the plane, so $\mathrm{CDOF} = 2 - 1 = 1$. The coordinate is x, whose motion law is given.
- **Coordinates:** x (and the arc length s along the curve).

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with the curve in the Oxy plane.
- **Parametric curve:** $\mathbf{r}(x) = x\,\mathbf{i}_0 + \sin x\,\mathbf{j}_0$, with $\dfrac{d\mathbf{r}}{dx} = \mathbf{i}_0 + \cos x\,\mathbf{j}_0$. The differential arc length is

$$ds = \left\| \frac{d\mathbf{r}}{dx} \right\| dx = \sqrt{1 + \cos^2 x}\, dx \equiv L(x)\, dx$$

- **Intrinsic basis.** $\mathbf{e}_t$ is the unit tangent and $\mathbf{e}_n$ points toward the centre of curvature (Notes Sec. 2.4). Both are obtained in 3.1. In the plane the normal can be written as the left normal $\mathbf{k}_0 \times \mathbf{e}_t$ up to a sign, because the curve has inflection points where the centre of curvature switches sides:

$$\mathbf{k}_0 \times \mathbf{e}_t = \frac{-\cos x\,\mathbf{i}_0 + \mathbf{j}_0}{L(x)}, \qquad \mathbf{e}_n = \operatorname{sgn}(\sin x)\,\frac{\cos x\,\mathbf{i}_0 - \mathbf{j}_0}{L(x)} \quad (\sin x \neq 0)$$

- **Change of basis.** The matrix with the components of $\mathbf{e}_t$, $\mathbf{e}_n$ and $\mathbf{e}_b$ in $\mathcal{B}_0$ is, for $\sin x \neq 0$,

$$[{}_0R_{\text{tnb}}] = \frac{1}{L}\begin{pmatrix} 1 & \operatorname{sgn}(\sin x)\cos x & 0 \\ \cos x & -\operatorname{sgn}(\sin x) & 0 \\ 0 & 0 & -L\operatorname{sgn}(\sin x) \end{pmatrix}, \qquad \det = \frac{1}{L^3}\left(-L\operatorname{sgn}(\sin x)\right)\left(-\operatorname{sgn}(\sin x) - \operatorname{sgn}(\sin x)\cos^2 x\right) = 1, \qquad RR^T = I$$

With $L^2 = 1 + \cos^2 x$ and $\operatorname{sgn}^2 = 1$, the rows have unit norm ($(1+\cos^2 x)/L^2 = 1$, $(\cos^2 x + 1)/L^2 = 1$, $L^2/L^2 = 1$) and are orthogonal (rows 1 and 2: $(\cos x - \operatorname{sgn}^2 \cos x)/L^2 = 0$; row 3 has no common nonzero entry with the others), so $RR^T = I$. For $\sin x > 0$, $\mathbf{e}_n$ is the right normal $-\mathbf{k}_0 \times \mathbf{e}_t$ and $\mathbf{e}_b = -\mathbf{k}_0$; for $\sin x < 0$ it is the left normal and $\mathbf{e}_b = \mathbf{k}_0$. In both cases the triad is right-handed.

Derivation

3.1 Kinematics

Velocity and acceleration

Pedagogical Justification: the position is expressed in the fixed basis, so the derivatives of the unit vectors vanish. With $\dot{x} = 1$ and $\ddot{x} = 0$ (chain rule $\dot{y} = \cos x\, \dot{x}$):

$$\mathbf{v} = \dot{x}\,\mathbf{i}_0 + \cos x\,\dot{x}\,\mathbf{j}_0 = \mathbf{i}_0 + \cos x\,\mathbf{j}_0, \qquad v(x) = \sqrt{1+\cos^2 x} = L(x)$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \ddot{x}\,\mathbf{i}_0 + \left(-\sin x\,\dot{x}^2 + \cos x\,\ddot{x}\right)\mathbf{j}_0 = 0\,\mathbf{i}_0 - \sin x\,\mathbf{j}_0 = -\sin x\,\mathbf{j}_0$$

(a) Intrinsic unit basis

- **Tangent:** $\mathbf{e}_t = \dfrac{d\mathbf{r}/dx}{\|d\mathbf{r}/dx\|} = \dfrac{\mathbf{i}_0 + \cos x\,\mathbf{j}_0}{L}$.
- **Pedagogical Justification (normal):** by the definition of the course, $d\mathbf{e}_t/ds = \mathbf{e}_n/\rho$, so $\mathbf{e}_n$ is the direction of $d\mathbf{e}_t/ds$. We differentiate with respect to x (product rule, with $\frac{d}{dx}L^{-1} = -\frac{1}{2}L^{-3}\left(-2\cos x \sin x\right) = \sin x \cos x\, L^{-3}$) and use $ds = L\,dx$:

$$\frac{d\mathbf{e}_t}{dx} = \frac{\sin x\cos x}{L^3}\left(\mathbf{i}_0 + \cos x\,\mathbf{j}_0\right) + \frac{1}{L}\left(-\sin x\,\mathbf{j}_0\right) = \frac{\sin x\cos x\,\mathbf{i}_0 + \left[\sin x\cos^2 x - \sin x\left(1+\cos^2 x\right)\right]\mathbf{j}_0}{L^3} = -\frac{\sin x}{L^3}\left(-\cos x\,\mathbf{i}_0 + \mathbf{j}_0\right)$$

$$\frac{d\mathbf{e}_t}{ds} = \frac{1}{L}\frac{d\mathbf{e}_t}{dx} = -\frac{\sin x}{L^4}\left(-\cos x\,\mathbf{i}_0 + \mathbf{j}_0\right)$$

- The norm of this vector is the curvature, and its direction is $\mathbf{e}_n$. The vector $-\cos x\,\mathbf{i}_0 + \mathbf{j}_0$ has norm L, so

$$\left\|\frac{d\mathbf{e}_t}{ds}\right\| = \frac{|\sin x|}{L^3} = \frac{1}{\rho}, \qquad \mathbf{e}_n = -\operatorname{sgn}(\sin x)\,\frac{-\cos x\,\mathbf{i}_0 + \mathbf{j}_0}{L} = \operatorname{sgn}(\sin x)\,\frac{\cos x\,\mathbf{i}_0 - \mathbf{j}_0}{L}$$

- For $0 < x < \pi$ ($\sin x > 0$) the arch is concave down and $\mathbf{e}_n$ points downward; for $\pi < x < 2\pi$ it points upward. At the inflection points $x = k\pi$ the curvature vanishes and $\mathbf{e}_n$ is not defined ($\rho \to \infty$).
- **Binormal:** $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = -\operatorname{sgn}(\sin x)\,\mathbf{k}_0$, constant in each arch.

(c) Tangent and normal acceleration components

> **Pedagogical Justification:** project the acceleration on the intrinsic vectors:

$$a_t = \mathbf{a}\cdot\mathbf{e}_t = (-\sin x\,\mathbf{j}_0)\cdot\frac{\mathbf{i}_0 + \cos x\,\mathbf{j}_0}{L} = -\frac{\sin x\cos x}{\sqrt{1+\cos^2 x}}$$

$$a_n = \mathbf{a}\cdot\mathbf{e}_n = (-\sin x\,\mathbf{j}_0)\cdot\operatorname{sgn}(\sin x)\,\frac{\cos x\,\mathbf{i}_0 - \mathbf{j}_0}{L} = \operatorname{sgn}(\sin x)\,\frac{\sin x}{L} = \frac{|\sin x|}{\sqrt{1+\cos^2 x}} \geq 0$$

- **Check:** $a_t^2 + a_n^2 = \dfrac{\sin^2 x\left(\cos^2 x + 1\right)}{1+\cos^2 x} = \sin^2 x = \|\mathbf{a}\|^2$.

3.2 Forces

Not applicable: kinematics-only problem. The curve and the motion law $\dot{x} = 1$ are given.

3.3 Equations of Motion

Not applicable: kinematics-only problem. The motion is prescribed by the statement.

3.4 Integration and Answers

(b) Arc-length parameter $s(x)$

> **Pedagogical Justification:** integrate the metric differential from $x = 0$ (taking $s(0) = 0$):

$$s(x) = \int_0^x \sqrt{1 + \cos^2 u}\, du$$

The integral cannot be written with elementary functions (extension: it is an incomplete elliptic integral of the second kind); the statement allows leaving it as an integral. Since the integrand satisfies $\sqrt{1 + \cos^2 u} \geq 1 > 0$, $s(x)$ is strictly increasing, so it can be inverted to give $x(s)$.

(d) Radius of curvature

> **Pedagogical Justification:** from $a_n = v^2/\rho$, with $v^2 = 1 + \cos^2 x$:

$$\rho(x) = \frac{v^2}{a_n} = \frac{1 + \cos^2 x}{|\sin x|/\sqrt{1 + \cos^2 x}} = \boxed{\frac{\left(1 + \cos^2 x\right)^{3/2}}{|\sin x|}}$$

As a check (extension), this is the standard curvature formula of a plane graph, $\rho = \dfrac{\left(1 + y'^2\right)^{3/2}}{|y''|}$, for $y' = \cos x$ and $y'' = -\sin x$.

Interpretation & Checks

- **Dimensional check:** $[v] = \mathrm{m/s}$, $[a_t] = [a_n] = \mathrm{m/s^2}$, $[\rho] = [v^2/a_n] = \mathrm{m}$ and $[s] = \left[\int \sqrt{1 + \cos^2 u}\, du\right] = \mathrm{m}$ (using the length scale 1 m inside the sine).

- **Special points:** at the inflection points $x = k\pi$, $a_n = 0$ and $\rho \to \infty$ (locally straight). At the crest $x = \pi/2$: $v = 1$ m/s, $a_t = 0$, $a_n = 1$ m/s^2 and $\rho = 1$ m, equal to $1/|y''| = 1$ m; there $\mathbf{e}_n = -\mathbf{j}_0$, toward the centre of curvature below the crest. At $x = \pi/4$: $\rho = (3/2)^{3/2}/(\sqrt{2}/2) = 3\sqrt{3}/2 = 2.598$ m.
- **Arc length:** $s(\pi) = \int_0^{\pi} \sqrt{1 + \cos^2 u}\, du = 3.8202$ m, greater than the horizontal distance π m, as it must be.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 6 in its solution keys, so the results are verified by independent derivation: sympy reproduces a_t, a_n, $a_t^2 + a_n^2 = \sin^2 x$, the curvature $\|d\mathbf{e}_t/ds\| = |\sin x|/L^3$ and $\rho(\pi/2) = 1$, $\rho(\pi/4) = 3\sqrt{3}/2$; numerical quadrature gives $s(\pi) = 3.82020$ m.

Source: Problems.pdf